

Governing AI Companionship: Lessons from Social Media Regulation

GUANYUE (YUMA) WU, University of Ottawa, Canada

Since the proliferation of AI companionship following COVID-19, the calls to regulate AI companion services have only grown in urgency. The attempted and actual suicides of several individuals, primarily children, have been linked to AI companions like ChatGPT and Character.AI, and there is also growing evidence that AI companionship enables addictive behaviours. This paper provides conceptual scaffolding for the task of regulating AI companions by connecting it to the task of regulating social media. By examining how regulators tackle the problems of harmful content and addictive designs on social media platforms, we can learn useful lessons for how the same problems can be tackled as they manifest in AI companionship. Our first lesson concerns the moderation of harmful content, where we find that social media regulatory measures like notice-and-takedown and age restrictions are applicable, though their efficacy is limited by the technology needed for enforcement. The application of intermediary immunity would require greater legal certainty via binding decisions or enforcement actions. The second lesson concerns restrictions on addictive designs, where we find that antitrust measures and litigation are likely to have a positive effect, though the current and potential impact of legislative measures remain uncertain. We conclude by identifying problems posed by AI companions – the accumulation of social power, and the macrosocial implications of proliferating AI companionship – that fall beyond social media regulation and necessitate novel solutions.

CCS Concepts: • **Social and professional topics** → **Governmental regulations; Antitrust and competition; Governmental regulations; Consumer products policy**; Technology and censorship; • **Applied computing** → *Law*; Psychology; Sociology; • **Human-centered computing** → Human computer interaction (HCI).

Additional Key Words and Phrases: AI Companions, AI Chatbots, AI Companions, Antitrust, Competition Law, Consumer Protection, Anthropomorphism, Emotional Attachment, Harmful Content, Addictive Behaviours, Dark Patterns, Addictive Design

ACM Reference Format:

GuanYue (Yuma) Wu. 2026. Governing AI Companionship: Lessons from Social Media Regulation. In *The 2026 ACM Conference on Fairness, Accountability, and Transparency (FAccT '26)*, June 25–28, 2026, Montreal, QC, Canada. ACM, New York, NY, USA, 22 pages. <https://doi.org/10.1145/3805689.3812294>

1 Introduction

A paradigm shift in human sociality began following the creation of the first chatbot, ELIZA, in the mid-60s [127]. ELIZA was programmed to identify keywords in users' natural language inputs and respond according to a conditional rules, such as by substituting "I" for "you" to turn statements into questions [127]. While primitive in hindsight, ELIZA was no less than the Mitochondrial Eve to today's colourful assemblage of commercial AI companions [127]. It pioneered the idea of socializing with computers by demonstrating that, so long as the computer elicited *just enough* associations with sociality, users would simply fill in the blanks [106, 127, 176]. There was no need for sentience, elusive as it is.

Despite this innovation, ELIZA and its descendants remained a niche curios until the global COVID-19 pandemic. Yearning for social connection amidst social distancing measures, people turned to specific-purpose companionship chatbots like Replika and general-purpose chatbots like ChatGPT, both of which had just received

Author's Contact Information: GuanYue (Yuma) Wu, yuma.wu@uottawa.ca, University of Ottawa, Ottawa, Ontario, Canada.



This work is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License.

FAccT '26, Montreal, QC, Canada

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2596-8/2026/06

<https://doi.org/10.1145/3805689.3812294>

a massive upgrade in coherency and usability following advancements in large language models [124]. Unlike humans, AI chatbots are always available and offer a judgement-free space for the user to explore taboo fantasies and sensitive topics [113, 141, 164]. In fact, research has shown that ad-hoc conversations with an AI chatbot can be on par with human interaction in relieving loneliness [61], though their effects are limited in alleviating isolation and improving social competency in the long term [113, 141, 184, 185]. Still, they were better than nothing. As word of the new technology spread, AI companion apps boomed in popularity. Starting with 1.5 million users in 2017, the Replika userbase ballooned by half a million new users in April 2020 *alone*, growing to 2 million total users by March 2023 [81, 124, 168].

Soon, however, the gloss of novelty began to wear thin. News reports of AI companions' potential for harm began cropping up around the world, raising questions surrounding safeguards and recourse. For instance, AI companions have generated harmful content that encouraged and romanticized suicide, which have been linked to the suicides of several people, including that of a Belgian man in 2023 and several US teens in 2024 and 2025 [51, 69, 152, 174]. Along the same lines, AI companions have been found to encourage disordered eating and self-harm [55, 70], as well as counselling users on criminal acts [167]. Users also develop unhealthy attachments to AI companions and patterns of compulsive engagement, which have been attributed to intentional addictive design patterns by the AI companion developers that aim to maximize engagement [44, 107, 115, 115, 136, 153]. As a result, regulators across jurisdictions such as the US, UK, and EU are turning their attention to AI companions, starting with requirements like AI disclaimers that regularly alert users to the artificiality of their conversation [60, 93]. However, these measures are limited in their efficacy, necessitating the search for other responses that strike closer to the heart of the risks posed by AI companions.

This paper provides a conceptual incision into the problem of AI companions by relating its harms to the harms posed by social media. We first define AI companions in Section 2. Then, we connect social media regulation to AI companion regulation in Section 3, motivating the lessons that we identify in Sections 4 and 5. In Section 4, we examine what social media regulation can teach us about mitigating harmful content in AI companions. We find that notice-and-takedown and age restrictions to be applicable to AI companions, though their efficacy is limited by the necessary technology. Intermediary immunity would only be applicable with greater legal certainty. In Section 5, we evaluate the applicability of measures curbing addictive design in social media to AI companions. We find that antitrust proceedings and litigation are likely to restrict addictive design, but the efficacy of legislation remain uncertain. To end, we explore questions that social media regulation leaves unanswered in Section 6 and conclude in Section 7.

2 What are AI Companions?

To start, this paper is broadly concerned with AI chatbots that enable repeated interactions and foster emotional attachment.¹ To facilitate the discussion, we can map the AI chatbots along two dimensions: purpose and embedding. Along the purpose dimension, there are *general-purpose* chatbots like ChatGPT and *specific-purpose* chatbots like Replika (designed for companionship purposes) and Harvey (designed for legal analysis and related work). As for embedding, chatbots can be *standalone* services like Claude, *embedded* in social media platforms like Snapchat's My AI, or *both* like Grok, which is available standalone and through X.² In this section, we will construct a definition of AI companions that circumscribes this 2D space in a way that align with our normative and functional concerns.

We take the normative position that the design and development of novel technology ought to be circumscribed by law only insofar as is necessary to prevent harm to user welfare. This has two implications. First, our analysis

¹Interactions with robots may also carry social implications, but robots pose distinct challenges that fall outside the scope and focus of this discussion.

²There are also chatbots that are embedded in other services like Gemini for Google. Because this discussion focuses on social media and AI companions, we will focus on ones embedded in social media platforms in our discussion of embedded AI chatbots.

is neutral towards the moral implications of relationships with AI companions. Our use of the word "companion" and "companionship" is intended neither to condone nor to reject social attachments to artificial entities. The second implication relates to the range of digital entities captured by our definition. Because we are primarily concerned with user outcomes, it follows that we are concerned with AI companions *as defined by users*. For instance, a 2025 study found that nearly a fifth of discussions on r/MyBoyfriendIsAI concerned ChatGPT, with other posts discussing relationships with Grok or Snapchat's My AI service [18, 141]. In other words, a satisfactory definition of AI companion for this discussion will not only capture specific-purpose chatbots designed to foster social responses, but ought to also capture other chatbots that users experience attachment to independent of purpose or embedding.

The functional purpose of a definition for AI companion in this discuss is to inform our discussion of regulatory measures. This also has two implications. First, the analysis of prospective regulatory measures is inherently forward-facing, which means that our definition must balance the need to be sufficiently specific about legally salient attributes of the technology with the need to be flexible towards future innovations. Therefore, our definition must capture the unique and relevant features of AI companions, but cannot be so specific that it precludes future iterations of the technology. Second, our functional goal necessitates balancing considerations of what users consider to be AI companions with the need for practical specifications. If our definition of AI companions solely centered on whether an user treated the digital artefact in a specific way, it would be too nebulous for our regulatory discussions. Instead, we need a definition with concrete technical specifications that implicitly captures users' relational experiences.

With these considerations in mind, we adopt a twofold definition: *an AI companion is an AI chatbot that engages in affective communication and affords the building of longitudinal relationships* [28]. Here, we use *affective communication* to refer to verbal and non-verbal interactions that evoke affective responses in users such as arousal and trust [28, 183]. Also, we use *affordances of longitudinal relationship-building* to refer to features that enable references to an identity and a shared history with the user. This definition meets our normative criteria because it is both purpose- and embedding-agnostic, instead focusing on the technology's welfare-augmenting abilities of communication and relationship-building. This definition also satisfies our functional criteria because it is not specific to current configurations of AI companions. Instead, this definition prescribes attributes that are neutral towards the underlying technology. Furthermore, this definition balances user experiences with practical considerations. By structuring our definition around the affordances of the technology, we circumscribe our focus on the technical capabilities associated with AI companionship rather than user perceptions, which enables recommendations of practical regulatory interventions.

3 Why learn from social media regulation?

In this section, we show that reasons for regulating social media resemble reasons for regulating AI companions, which indicates that the regulatory solutions may be transferable from the former to the latter. We begin with background on social media regulation before examining the two points of similarity: harmful content and addictive design.

3.1 Social Media Regulation

Whereas *online platforms* generally enable user interactions, *social media platforms* do so with the specific purpose of driving social engagement through the production and dissemination of user-generated content such as text or video [35, 88, 179]. In the 90s, internet media was predominantly disseminated following the one-to-many model of traditional broadcast stations and publishing houses [35, 179]. That changed in the 00s with the rapid emergence of social media platforms like Facebook and YouTube, which blurred the line between

consumers and producers into a web of many-to-many interactions between users around the world, flooding the internet with user-generated content [179].

As is the case for AI companions, serious concerns about the potential for harm bubbled up following social media's meteoric rise. We can observe three potential causes of harm here: *harmful content*, *addictive design* patterns, and *privacy violations*. First, realizing they were unbound by professional standards and shielded by anonymity or pseudoanonymity, some users began sharing *illegal and harmful content* on social media platforms. Some prominent examples of such content include child sexual abuse material [57, 132], misinformation pertaining to COVID-19 and recent US elections [83, 143, 161], and content promoting self-harm and disordered eating [59, 123, 144]. Second, by optimizing for user engagement, social media platforms deployed *addictive design* features like infinite scrolling that became associated with patterns of dependency and addiction [32, 48, 148]. A 2018 survey found that 6.85% of Canadian children experienced at least 6 of 9 problematic social media behaviours such as feeling uncomfortable when deprived of social media access, while 33.14% reported 2-5 [46, 173]. Third and finally, users were alerted to misappropriation of the personal data following the 2018 Cambridge Analytica scandal, when news broke that the firm illegally collected data from at least 87 million Facebook users to derive insights in support of the Trump campaign [39, 120, 179]. While important, privacy concerns fall outside the scope of this discussion because the mechanisms for safeguarding user privacy are generally neutral towards social media platforms and AI companions. Unlike the other safeguards that we will discuss, privacy safeguards for social media *already* apply to AI companions, so it is redundant to discuss them here. Instead, we will be focusing on what we can learn from regulatory attempts to mitigate the first two concerns.

3.2 Harmful Content in Social Media and AI Companions

First, like the objections to illegal and harmful content on social media, many objections to AI companions can be framed as objections to harmful content. Government and private parties alike have objected to the presence of illegal and harmful content on both social media and in interactions with AI companions.

To start, government objections with respect to harmful content on social media and AI companions are becoming intertwined. Several policy proposals in multiple jurisdictions have discussed the two technologies in conjunction, including the UK government's public consultations on digital well being, which covers both matters pertaining to harmful content and addictive design (see more in the next section) with respect to both social media and AI chatbots [100]. Similarly, discussions surrounding online safety in Canada have also encompassed social media and AI chatbots in the same breath, citing parallel concerns regarding harmful content with a particular emphasis on protecting children [108, 165]. Indeed, the overarching policy motivations cited for regulating social media and AI companions carry a harmonious undertone: concerns about children's online activities and the disproportionate harms that malicious online actors may have on their well-being. Another component of this overlap in government attention is rooted in the increasingly blurring lines between social media and AI companions as large social media platforms like Meta and Snapchat feature AI chatbots in increasing prominence [17, 131]. Therefore, depending on the degree to which these embedded chatbots operate as an extension of the social media platforms and constitute AI companions under our definition, it may already be the case that certain social media regulations *already* apply to AI companions.

Legal proceedings initiated by private parties regarding harmful content against social media platforms and AI companions also bear close similarities. Like the social media platforms that have been sued for contributing to the death and mental disorders of several people (disproportionately teenagers) [22], AI companions have been alleged to contribute to the suicide of several individuals (also disproportionately teenagers) in ongoing legal proceedings in the US [51, 69, 152]. Both sets of lawsuits impugn the availability of harmful content, such as media promoting disordered eating and self-harm [22, 181], as well as age-inappropriate content available to young users [21, 56, 69, 152].

3.3 Addictive Design in Social Media and AI Companions

Second, concerns of emotional dependency to AI companions echo concerns of addiction to social media.

Several lawsuits have been brought against social media platforms for this issue, arguing that social media platforms optimize for "compulsive use" through design features such as infinite scroll, auto-playing videos, and notifications that induce fears of missing out [32, 68, 112]. As for AI companions, there is emerging evidence that users exhibit behaviours associated with addiction, and some report feeling addicted to and emotionally dependent on the constant availability and relentless emotional support of their AI companions [107, 113, 115, 119]. In particular, researchers point to design elements of AI companions that may invite these addictive behaviours, including variable rewards like always-agreeable responses and notifications that, like in social media, pull the user back into the app [136, 153, 178]. Furthermore, some AI companion services also integrate gamification features from free-to-play gaming such as daily rewards and unlockable content to bring users back to the application day after day [85].

A further similarity is the motivations underlying their addictive design elements – engagement-based business models [85, 87, 133, 148]. For both social media and AI companion companies, more engagement is better. While many social media platforms profit directly from means of engagement such as subscription services for premium treatment like custom emoticons and ad-hoc in-app purchases that enable different forms of engagement (e.g., Reddit awards), engagement is monetized predominantly indirectly through the sale of advertisement placements and user data [45, 148, 180]. The more time social media users spend on a platform, the more opportunities they have to generate valuable usage data and encounter advertisements, in turn creating more monetization opportunities for the social media companies [148]. Similarly, AI companion companies are also keen to capitalize on user engagement as social media platforms have done. For instance, several AI companion services also offer subscription services that, like their social media counterparts, promise to enrich user experience through exclusive features such as additional modalities of interaction [85, 90]. Some AI companions also offer ad-hoc purchases such as cosmetic options for the companion and additional interactions with AI companions beyond arbitrary daily interaction caps [85, 183]. While advertising in AI companions have only just started to roll out this year with ChatGPT, it is likely only the beginning of ad integration in AI companions [101, 147, 172]. As is the case for social media, advertising profits create a profit incentive to maximize engagement with AI companions as engagement grows proportionately to ad placement opportunities, which in turn strengthens the profit incentive and the pursuit of greater engagement [148].

Given these similarities, we can examine the solutions proposed for social media and apply them to AI companions.

4 Lesson #1: Harmful Content and the Health of the Public Sphere

4.1 Solutions to Harmful Content in Social Media Regulation

4.1.1 Challenges in Regulating Harmful Content in Social Media. To begin, it is difficult to construct a *definition of harmful content* that is sufficiently rigid and detailed as to be applied in platform moderation while still remaining flexible and expansive as to accommodate differences in culture and norms [102, 157]. An overly narrow definition draws concerns that some harmful content may not be captured, while an overly expansive definition raises the concern of *censorship* [102]. A further complication is the *categorization* of harmful content – it would be inefficient to treat rude comments with the same urgency as child sexual abuse content, for instance [157]. In particular, some content is safe for the general public but poses additional risks to young users, necessitating context-sensitive distinctions.

Besides definitional issues, there remains the sheer scale of social media. Facebook boasted 2.85 billion active users in 2021, with 1.88 billion of those users checking in daily [163]. This poses significant *cost challenges* as platforms balance user engagement and legal obligations with the cost of moderation [58, 72, 102]. Not only do

platforms contend with costs of developing automated moderation processes for a deluge of media, but they also employ an army of human moderators to review removal decisions [58, 102]. Furthermore, the scale of social media also poses *enforcement challenges*. Because harmful content may originate from anonymous or pseudonymous users around the world, the resources required for government intervention or litigation are transnational, complex, and costly.

4.1.2 Current Social Media Regulation. Proposed solutions to these challenges include intermediary immunity framework, notice-and-takedown regimes, and age restrictions. We will examine these exemplary approaches below.

In the US, platforms enjoy *intermediary immunity* for harmful user-generated content under Section 230 of the *Communications Decency Act* ("Section 230") [43, 102, 111]. Prior to Section 230, platforms attracted liability for user-generated content if they attempted to moderate content because it made them *publishers* who are responsible for the content they display by virtue of their editorial-like control, and not mere *distributors*, who are generally not responsible for such content [102, 111]. And yet, platform providers *want* to moderate – harmful content erodes user engagement, economic interests, and public relations [102, 111]. Section 230 ensured that platforms would not be considered publishers *even if* they engaged in content moderation, shielding them from liabilities for the content they hosted [43, 88, 102, 111]. The 2022 EU *Digital Service Act* ("DSA") Article 6 provides similar immunity for "hosting" services that store user information such as social media platforms, whereby the service is exempt from liability for illegal content submitted by third parties provided that they lack "actual knowledge" of the illegality [8]. Like Section 230, DSA Article 7 provides that voluntary moderation efforts do not vitiate the Article 6 immunity [9]. Unlike Section 230, the DSA qualifies this immunity in two ways. First is the aforementioned "actual knowledge" exception, which we will explore in the next paragraph. Second, the DSA imposes additional obligations on "Very Large Online Platforms" (VLOPs) with more than 45 million average monthly users in the EU, including actively monitoring for "*systemic risks*" such as illegal and harmful content [4, 7]. The advantage of intermediary immunity is that it relieves regulators of enforcement burdens and enables the parties with the most control over platform content – the platforms – to handle the task. However, Democrats argue that Section 230 fails to motivate sufficient moderation, while Republicans argue that it protects platforms practicing censorship [111]. These censorship concerns also follow the DSA – VLOP obligations encourage content control beyond what is strictly illegal, and the varying designations of illegal content across EU states may lead to a oppressive web of criss-crossed restrictions that bind all VLOP users, EU or not [169].

Notice-and-takedown is another regulatory response to harmful content. These measures require platforms to remove (i.e. *takedown*) illegal content from general access following receipt of a verified *notice* of impugned content on that platform [29, 105]. Under the Article 16 of the EU DSA, social media platforms are required to provide mechanisms for individuals or other entities to report illegal information on the platform, such as "Report" buttons attached to posts [5]. Article 9 also obligates removal following orders from national authorities [10]. These two forms of notifications fulfill the "actual knowledge" requirement under Article 6, which in turn obligates the platform to "expeditiously to remove or to disable access to the illegal content" [8]. Similarly, the US also maintains "bottom-line" obligations regarding the removal of illegal content such as non-consensually published intimate images [92, 162]. Under US regimes, failing to remove content exposes the provider to legal liability, such as criminal liability where the content is illegal or regulatory liability as enforced by agencies like the Federal Trade Commission ("FTC") [105, 162]. Notice-and-takedown regimes mitigate harmful content to the extent that there are motivated parties to file takedown notices and platforms that promptly comply. In both the US and EU, this entails a mix of government and private actors. Where private actors are active in reporting, notice-and-takedown measures can alleviate the enforcement pressure from government bodies and the cost

of detecting harmful content from online platforms. However, a lack of a unified standard for harmful content among private parties raises the potential for private censorship and frivolous complaints.

Another nascent response to harmful content is *age-based restrictions* on content and platforms. In 2024, the UK *Online Safety Act* ("OSA") came into force, which imposed requirements on platforms to implement age-based restrictions on content and access [24, 109, 134]. Similarly, the DSA requires VLOPs to "taking targeted measures to protect the rights of the child, including age verification" [3]. The reasoning is that children are more vulnerable than adults to harmful and age-inappropriate content such as pornography, necessitating either bans that bar children from platforms or restrictions on access to age-inappropriate content [73, 159]. These age verification measures are politically appealing and sidestep complexities pertaining to definitional issues and censorship [77]. However, the rollout of age assurance requirements has encountered no shortage of barriers. We have previously discussed the difficulties relating to identifying and removing harmful content for some or all users, but age verification also poses a further problem: how does one verify an user's age? Unfortunately, existing technical measures have proven to be unreliable at best and privacy-endangering at worst [34, 74, 99]. Early research not only indicates weak performance in estimating the age of children through facial recognition, but also finds that the error rates are higher for individuals of African descent compared to those of Caucasian descent [160]. As for privacy implications, recent data breaches of age verification information have raised concerns – the social messaging platform Discord experienced a data breach in 2025, leading to the exposure of over 70,000 images used for age verification, many of which were images of government ID [53, 86].

4.2 Analysis: Applying Social Media Solutions to AI Companionship

4.2.1 Challenges in Regulating Harmful Content in AI Companions. AI companion regulators face similar definitional issues as social media regulators in their attempts to pin down what is "harmful" about AI companion outputs. As is the case for social media harms, different types of content can pose different harms to different people, and attempts to curb select types of content inevitably draw concerns relating to paternalism and censorship. However, *unlike* in social media, the censorship concerns are more nuanced. When an AI companion is prohibited from generating some content, there is no human expression being directly censored. However, one could argue that the user, who is asking the AI companion to generate that content, is being censored *indirectly*. In a sense, content restrictions raise concerns that AI companion providers are denying types of artificial relationships that are available [119, 139, 141].

Compared to Facebook's 2.85 billion, AI companions have a much smaller user base. ChatGPT, the largest AI companion by our definition, only reached 900 million users in March 2026 [117]. The user base is even smaller for specific-purpose AI companions – Replika's 30 million and Character.AI's 20 million are tiny by comparison [141]. However, even these smaller user pools pose scale-related problems. In theory, the cost of detecting harmful content ought to be lower for AI companions because the platforms control the "source" of harmful content in question – the Large Language Models (LLMs) responsible for generating text and media in response to user inputs. All AI companions today use LLMs, though each company may customize its models for specific purposes [38]. Unfortunately, the technical challenges of detecting harmful content in LLM outputs have proven difficult to solve. Replika's founder remarked that, despite advancements in AI moderation, "you just can't account for every single possible thing" that users may seek to generate [54, 67]. That said, because the harmful content is coming from the AI companion service rather than a constellation of users, enforcement and liability is easier to realize. There are several lawsuits against AI companions for harmful content, as well as fines from the European Data Protection Board [36, 69, 152].

4.2.2 Learning from Current Social Media Regulation. Here, we will analyze whether each of the three strategies – intermediary immunity, notice-and-takedown, and age restrictions – would be appropriate and effective.

First, intermediary immunity under the DSA is likely applicable to AI companions. Social media platforms are captured by the DSA [8], meaning that embedded AI companions likely fall under the same purview. As for standalone AI companions like ChatGPT and Replika, they store user information such as conversation logs that enable longitudinal relationships, meaning that they are also likely "hosting" intermediaries under the DSA. Notably, the applicability of VLOP obligations cuts across our definition of AI companions. On one hand, no standalone AI companion providers has been designated a VLOP³ under the DSA at this time [19]. On the other hand, embedded AI companions like Meta's chatbots are captured by virtue of their parent platforms' VLOP designations [19]. This may lead to shifting obligations as AI companion services become VLOPs, as well as stricter obligations on embedded-type AI companions. The application of intermediary liability in the US requires further analysis. Section 230 would only apply if AI companions are "*content neutral tools*" like search engines, where the service provided is agnostic to the substance of the content, as opposed to services that "create or develop outputs absent any prompting" [31, 128, 149]. It follows that online services designed to retrieve "third-party input", verbatim or summarized, will likely enjoy Section 230 immunity [31, 93].⁴ By this standard, AI companions are unlikely to be entirely immune. One of the main draws of AI companions is that they can create bespoke responses to users that are, by definition, distinct from existing content. Furthermore, some AI companions like Nomi.AI proactively message users with unprompted generated content [89], which fall outside of Section 230 protections. Without further litigation or legislative clarification, it is unclear whether intermediary immunity would be applicable. That said, what if intermediary immunity did apply in the US, as it likely does in the EU? Like social media platforms, AI companion services are motivated to restrict harmful outputs insofar as they pose a threat to their business. OpenAI has introduced several safeguards to ChatGPT, and smaller players like Character.AI have also introduced safeguards following legal pressure and public outcry [52, 122, 138]. Therefore, offering AI companion services intermediary immunity may remove a legal barrier to content moderation in the same way as it did for social media platforms, but the drawbacks are similar as well. Intermediary immunity for AI companions may unjustly shield services that permit the proliferation of harmful content, depriving harmed users of potential remedies.

Second, some AI companions like Character.AI have implemented notice-and-takedown procedures, though the focus appears to be avoiding copyright infringement [52]. In theory, a notice-and-takedown regime for AI companions would enable users to identify harmful outputs that the providers could then remove. In particular, because AI companions likely fall under DSA's definition of a hosting service, it is likely that these obligations already apply to AI companions, though there have yet to be any enforcement actions on the basis of the DSA [8]. Extending these obligations to AI companions could alleviate the lofty costs of content moderation and contribute to greater user agency in their relationships with their AI companions. Unfortunately, AI companion providers' responses to harmful content have not been promising due to technical challenges relating to LLMs [54, 67, 122, 170]. For instance, consider the challenge of removing harmful content from a LLM. Unlike harmful content on social media, which the platform can delete or remove access to, the same task is far more difficult for LLMs [93]. A basic issue is that current LLMs cannot distinguish between the contexts that make something harmful [98, 183]. Exemplifying this difficulty, the last conversation between Sewell Setzer III and his Character.AI chatbot could be considered benign roleplay without the context that he killed himself immediately afterwards. "What if I told you I could come home right now?" Setzer writes [150]. The chatbot responds encouragingly: "Please do, my sweet king [150]." Given these circumstances, notice-and-takedown is ill-suited to regulating AI companions given the current state of LLMs, though it may become feasible as the technology improves.

Third, age restrictions are the most popular self-regulatory response from AI companion services at this time. Following lawsuits alleging that Character.AI contributed to the deaths and attempted suicides of four

³ ...or a Very Large Online Search Engine (VLOSE), which bears the same obligations as VLOPs.

⁴For an in-depth analysis of whether Section 230 applies to LLMs, please see "Section 230 Won't Protect ChatGPT" by Matt Perault [142].

teens [51, 122, 150, 152], the platform released both content-based and access-based restrictions, where users under 18 are limited to two hours of daily use with restrictions on "unsuitable" content [79, 103].⁵ The platform also introduced the Parental Insights feature, which sends guardians summaries of their child's activities [79, 80]. Furthermore, the DSA's VLOP obligations encompass AI companions embedded in VLOP platforms, meaning that such AI companions are required to implement age restrictions by EU law [3]. Unfortunately, these age restrictions for AI companions have many of the same limitations as age restrictions for social media [67, 79, 103]. Where the restriction is *access-based* – limiting access to the service – AI companions also suffer the technical challenge of effective age verification [77, 79]. Where the restriction is *content-based* – limiting the kinds of available content – not only does the age verification challenge still apply, but services must also contend with the detection and removal issues that plague the notice-and-takedown solution. That said, supervisory features like Parental Insights can be helpful as they enlist parents to detect problematic use, offloading some of the supervisory costs from the platform. In short, age restrictions may be part of the regulatory solution, especially as the age verification technology improves, but its efficacy and scope are limited.

In short, there is no silver bullet. Extending intermediary immunity to AI companions would remove legal barriers to enacting content controls, but it does not motivate moderation on its own. Age restriction and notice-and-takedown measures would enable the protection of young users and crowdsourced content control respectively only if age detection and harmful content detection/removal technology for LLMs improve.

5 Lesson #2: Addiction by Design and Regulation by Design

5.1 Solutions to Addiction by Design in Social Media Regulation

5.1.1 Challenges in Regulating Addiction by Design in Social Media. The problem of addictive social media is situated in the wider problem of the proliferation of *dark patterns*, which the *California Privacy Rights Act* (CPRA) defines as "user interface[s] designed or manipulated with the substantial effect of subverting or impairing user autonomy" [42, 133, 148]. For the purposes of this discussion, we use *addictive design* (also known as "hyper-engaging dark patterns") to refer to dark patterns that specifically aim to induce user dependency and compulsive engagement [76, 153]. Examples of such designs include gamification through arbitrary achievements and variable reward signals like irregular notifications [78, 129].

The first challenge, as was the case for harmful content, is to provide a *definition of addictive design* with sufficient detail as to "form the basis for regulation while remaining flexible enough to accommodate new forms of addictive digital behaviour" [48]. First, there is a *threshold problem*: when does an engaging user pattern cross into the addictive design territory? For instance, notifying users about new videos can be helpful. However, when combined with other features such as login rewards, notifications contribute to an overall user experience where they feel constantly "pulled back" into the app [78, 129]. This is especially concerning given the "bidirectional relationship" between social media addiction and mental illnesses, where a negative feedback loop between worsening mental health and increasingly problematic levels of social media consumption may occur [114]. Therefore, whether a feature is addictive depends not only on the behavioural outcomes, but also on other contextual attributes that populate the user experience. This complexity precludes straightforward prohibitions and instead necessitates holistic analysis of digital experiences.

The second challenge of addiction by design is that, unlike harmful content, social media platforms have *no intrinsic motivation* to solve the problem even if it was well-defined. In fact, they are motivated to do the opposite. The proliferation of harmful content is disadvantageous to social media platforms: damage to public reputation, degradation of user experience, and shrinkage of the user base, all of which strike at the underlying business model of social media – the monetization of user engagement. In contrast, the compulsive use patterns associated with addictive designs align perfectly with the engagement model [133, 148]. The more addictive the

⁵The age verification mechanism employed by Character.AI has yet to roll out at the time of writing [103].

platform is, the more engaged users become, which then generates more monetization opportunities and creates more profit [133, 148]. As a result, addictive design cannot be curbed by measures that rely upon voluntary actions by platforms in the same way that intermediary immunity and notice-and-takedown rely upon the active cooperation of platforms [62, 148].

5.1.2 Current Social Media Regulation. As with harmful content, we will examine some exemplary approaches, including antitrust/competition law and litigation in the US and Canada, and direct regulation in the EU.

One of the first government responses to addictive design elements is the application of *antitrust and competition frameworks* in the US and Canada, respectively. In the US, the *Federal Trade Commission Act* ("FTC Act") grants the FTC broad powers to prevent any "representation, omission or practice that is likely to mislead the consumer" [1, 91]. In 2020, this authority was expanded to include the prevention of dark patterns, which the FTC described as interface elements that are "designed to subvert [consumer] autonomy" [118, 125]. While this expansion resulted in enforcement actions against several platforms, there has yet to be one focused on addictive design [116, 130]. In Canada, the *Competition Act* grants the federal Competition Bureau similar powers with respect to "representation[s] to the public that [are] false or misleading" [2]. In 2024, this authority was also expanded to the prevention of dark patterns, but there have been no enforcement actions on this basis as of yet [126]. The advantage of this approach is that the necessary administrative and enforcement frameworks are already in place. By expanding existing authority, enforcement agencies can leverage existing institutional resources towards evaluating platforms, which in turn informs both potential enforcement actions and future legislation. Furthermore, guidance issued by enforcement agencies can deter platforms from adopting design patterns even without concrete enforcement actions. Unfortunately, the applicability of antitrust and competition law to social media is limited. Unlike traditional business models, engagement-based businesses offer their services for free to the users, instead deriving profit from through transactions with third parties like advertisers [133, 148]. This leaves most actions of social media platforms under the radar of antitrust and competition authorities [118].

Although *litigation* is not a structured, government regulatory response, it is a powerful vanguard for regulating addiction by design on platforms in the absence of legislation. As discussed, social media platforms are alleged to deploy addictive design patterns in several lawsuits in both Canada and the US [32, 68, 112]. These lawsuits are powerful, as they often feature compelling user impact narratives that raise awareness of potential harms, leading to public pressure on the defendant firm to mitigate harm even before trial [67, 68, 122]. Furthermore, such lawsuits often feature sophisticated plaintiffs, which increases the probability that the platforms will be compelled to disclose informative insights on their operations [49, 68, 94, 112, 135]. Mounting lawsuits against social media platform regarding the use of addictive patterns address both definitional and motivational challenges. First, they provide courts with the opportunity to articulate a legally binding definition that will inform future enforcement and litigation across jurisdictions. Second, lawsuits highlight the causal link between the engagement-based business model of the platforms and potential harms. This brings awareness to the potential harm of the engagement model, exerts public pressure on the platforms to remove the impugned designs, and, if the plaintiffs succeed, would temper platforms' pursuit of further user engagement with legal caution. However, such lawsuits against large tech companies have historically ended in settlements, if at all [30, 146, 158]. While settlements may still result in improvements to online platforms, they often preclude the publication of a full ruling, which would contain useful information and binding guidance [75, 145]. In short, settlements may mitigate harms, but they do not always inform future regulations nor strike at the underlying interests at play.

In contrast to Canada and the US's reluctance to target addictive design with *legislative acts*, the EU has actively legislated on this subject. Starting in 2022, the EU has attempted to subsume dark patterns under the wider umbrellas of EU's 2005 *Unfair Commercial Practices Directive's* (UCPD) prohibition on "misleading" and "aggressive" commercial practices and 2015 *General Data Protection Regulation's* (GDPR) transparent data processing requirements under Article 5(1)(a) by issuing guidance that include dark patterns under the consideration of

these provisions. [4, 47, 71]. These efforts ramped up starting in 2022 with the DSA, when the EU began to directly address dark patterns by prohibiting interfaces that "impair [users' capacities] to make autonomous and informed choices or decisions" under Article 25 [6, 11]. Recital 67 identifies these interfaces as "dark patterns", listing examples such as undue friction in attempts to disengage [11]. In particular, Recital 83 recognizes addictive designs as systemic risks that VLOPs must mitigate [3, 12]. This obligation is exemplified by the first enforcement action under the DSA against TikTok (a VLOP) for its TikTok Lite Rewards programme, which the Commission argued "may stimulate behavioural addictions" in contravention of TikTok's obligations to mitigate systemic risks [26, 104]. While the 2024 EU *Artificial Intelligence Act* ("AI Act") does not use the term "dark patterns" or "addiction", it shares similar prohibitions on "purposefully manipulative or deceptive techniques" that impair users' abilities to make an informed decision using AI systems [13, 14]. Most recently, the EU Commission has proposed a new legislative act, the *Digital Fairness Act* ("DFA") [121], which seeks (among other objectives) to curb addictive designs, defining them as interface elements that actively encourage greater engagement and/or monetary investment, such as gambling-like rewards in online gaming [121]. The main benefit to the EU's proactive legislation is its precautionary nature – these legislative responses and guidance rapidly imposed structures and obligations on the previously unchecked sphere of interface design, which has resulted in informative insights on the efficacy of such rules. That said, the various definitions in these efforts are not harmonized, leading to under-enforcement in the gaps between the definitions and inconsistent enforcement where the definitions clash [42, 47, 110]. Furthermore, in part because of the preemptive nature of the efforts, the task of refining and updating the definitions as the technology changes will be a difficult and ongoing one [42, 47].

5.2 Analysis: Applying Social Media Solutions to AI Companionship

5.2.1 Challenges in Regulating Addiction by Design in AI Companions. Scholars have identified elements of addictive design in AI companions that not only echo the addictive design found in social media, but also resemble design elements from online gambling and gaming that foster similar addictive behaviour patterns [20, 48, 85, 115, 136, 153, 178]. This leads to similar challenges as regulating addictive patterns in social media, as well as some interesting differences.

To start, AI companions draw the same definitional issues as addictive design on social media. When is an AI companion engaging to the point of addiction? For instance, ChatGPT's ability to recall an user's past queries can be both useful and engaging – it can foster user trust and self-esteem by validating their concerns [136, 153]. However, as users build an expectation that ChatGPT is trustworthy and validating, it can generate a "feedback loop of validation" that raises the risk of problematic use, such as where the user feels anxious without the chatbot and thereby compelled to continue using it [113, 136]. Like the bidirectional relationship between social media addiction and mental illnesses, recent research suggests that individuals who feel isolated are prone to problematic use, which also poses the risk of a negative cycle [113, 114]. As is the case for social media, whether an AI companion fosters addictive behaviours depends on both the design features of the AI companion and user attributes.

AI companions also align closely with social media in terms of the motivational challenge, though the avenues for monetization are slightly different. Unlike most social media platforms, few AI companion applications have inserted advertisements in the user experience because of the significant potential to disrupt the user experience and reduce engagement. Most AI companion platforms like Replika and Nomi.AI instead offer subscription services that upgrade the user experience [85, 156, 183]. However, most AI companions still offer a free tier, and given that 95% of Replika's users only use the free plan, it stands to reason that the underlying business model is likely still an engagement-focused one [156]. Furthermore, many of the user experience upgrades for AI companions appear to motivate further engagement – additional generated content, additional modalities for interaction, to name a few [85, 183]. Therefore, like social media platforms, measures to limit user engagement

are likely to be antagonistic towards AI companion providers' business interests, meaning that self-regulatory measures are unlikely to be successful.

5.2.2 Learning from Existing Social Media Regulation. In contrast to the problem of harmful content, the differences between solutions for addictive design in social media versus AI companions are smaller. Here, we will apply the antitrust/competition law, litigation, and legislative frameworks to AI companions.

Unlike social media platforms, AI companions have not yet attracted the same level of scrutiny from antitrust and competition authorities, but that may soon change. In September 2025, the FTC issued an investigative order to seven AI companion companies for information on how they "measure, test, and monitor potentially negative impacts of this technology on children and teens", which includes emotional dependency and addictive behaviour [66, 82, 166]. The results of the investigation will likely inform legislative efforts and enforcement actions where warranted [82, 166]. This inquiry "signals a significant increase in regulatory scrutiny of AI" by the FTC, indicating the FTC's intention of shaping AI companion policy [166].⁶ The order also suggests that the application of antitrust law to regulating AI companionship will enjoy the same advantages as with social media. Like the FTC enforcement actions against social media platforms, the FTC order put AI companion providers on notice for potential enforcement and may deter AI companion designers from introducing further addictive designs [82]. Unlike the social media enforcement actions, which lacked clear precedent, antitrust authorities can rely on past arguments against social media in their enforcement actions against AI companions. In addition, significant players in the AI companion sphere are also social media companies (e.g., Meta), so enforcement actions in the social media sphere will also contribute to regulating AI companions [166]. However, the drawback that antitrust and competition law focus on traditional business models still applies. Overall though, the net effect of applying antitrust/competition law to AI companions is likely to be positive.

AI companions are named in a comparable number of lawsuits to social media. While most are concerned with harmful content [51, 122, 150, 152, 181], there is one that touches on their addictive nature: the lawsuit against Character.AI by Sewell Setzer III's mother, Megan Garcia, where she alleges that Setzer became addicted to the app as a result of "inherently dangerous features intended to deceive consumers and ensure dependencies" [65, 84, 152]. Pressured by the lawsuit and the news coverage, Character.AI implemented safety measures but notably did not directly address the issue of addictive design [51, 65, 67, 150, 152]. That said, the lawsuit is still ongoing, joined by a growing group of *amicus* from industry and academia, signalling burgeoning scholarly interest and an increasing body of valuable discussions [84]. Like the social media lawsuits, Garcia's lawsuit may result in a settlement, but it has already drawn significant regulatory attention to the link between AI companions and addictive behaviours. In fact, because AI companions are not as established as social media, lawsuits and public opinion exert greater pressure on the providers to make improvements. In sum, lawsuits are an effective means of combating addictive design in AI companions.

Finally, while there is yet to be any legislative deliberation targeting AI companions specifically, legislation aimed at social media and AI chatbots generally are likely to apply variably. We will address each in their own paragraph.

To begin, the UCPD and GDPR are likely to extend to addictive design patterns in AI companions. The UCPD could potentially extend to misleading and aggressive behaviour of AI companions that aim to keep users on the platform, and European Data Protection Agencies may bring enforcement actions for features of AI companion services that fail to adhere to GDPR principles. Notably, the Italian Data Protection Agency ("the Garante") is perhaps the most active enforcement authority in the AI companion space as of 2025. The Garante has brought several enforcement actions against AI companions like ChatGPT and Replika in 2023 and DeepSeek in 2025, resulting in steep fines – 5 million Euros for the parent company of Replika, and 15 million Euros against OpenAI [25, 33, 36, 182].

⁶There is no comparable action from the Canadian Competition Bureau.

The DSA is also likely to be applicable. DSA Article 25's general prohibition of designs impairing user autonomy is likely to apply to all AI companions, since all AI companions are either hosting services or embedded in one. Because of the definitional complexity of addictive patterns and the absence of enforcement actions pursuant to the DSA, it remains to be seen precisely what configurations of AI companions would be prohibited. For instance, hiding the option to downgrade one's paid subscription is likely within the scope of Article 25 prohibitions, but what about an AI companion that calls the user a cheapskate for downgrading their service [38]? Similar uncertainties entangle VLOP obligations. If an AI companion is designated as a VLOP and features elements like the TikTok Lite Reward, EU authorities could argue that the AI companions pose a systemic risk, but what about AI companions that engender emotional dependence through constant availability and unfailingly positive attitudes [104, 153]? More information derived through further guidance and enforcement actions will be needed to answer these difficult questions.

Like the DSA, the AI Act also accords different obligations to different AI companions [16]. General-purpose AI companions are General Purpose AI Systems ("GPAISs"), meaning that they bear transparency obligations such as maintaining technical documentation of its model in addition to the general prohibition on manipulation or deception [14, 16]. While transparency may enable accountability and deter intentional wrongdoing, it does not address addictive design directly. Specific-purpose AI companions are not GPAISs, but limited risk systems alongside other specific-purpose chatbots [16]. The main obligation of these limited risk systems is *content labeling* – users must be made aware that they are interacting with an AI system, and generated outputs must be labelled as such [15]. Unfortunately, content labeling alone cannot curb emotional dependency or compulsive use – user discussions indicate that disclaimers of AI system responses and other content labeling measures do little to dampen users' attachments [64, 141, 183].

Finally, given its objective to safeguard online activities, the DFA will likely cover large-scale standalone AI companions like ChatGPT and AI companions embedded in social media platforms at minimum, if not also the smaller-scale specific-purpose AI companions [121]. Based on the DFA's definition of addictive designs, gamification features analogous to existing gambling-like rewards such as loot boxes will likely be prohibited. However, like the DSA, it remains to be seen whether the DFA will seek to restrict chatbot-specific manipulations [38, 183].

Legislating against addictive designs in AI companions share the same advantages and disadvantages as with social media. As AI companion technology developed in past years, these laws remain the most relevant to the matter by virtue of their precautionary nature. Even in the absence of enforcement against AI companions at this time, these laws nevertheless impose structure and obligations in an otherwise unregulated space, which may also deter the use of potentially addictive designs generally. However, these laws have also resulted in a complex web of cross-cutting obligations, resulting in a legislative scheme that may not be harmonized towards an overarching objective. For instance, a specific-purpose AI companion that is considered a VLOP under the DSA may not be subject to transparency requirements under the AI Act, though it shares much of the underlying risks as a general-purpose AI companion. Ultimately, the application of these measures will only become clear with future enforcement actions and guidance.

As was the case for harmful content, there is no single best solution. Antitrust and competition law can play an informative and deterrent role, but the responsible authorities are restricted in their enforcement capabilities. Litigation is informative and can result in concrete penalties and remedies, but its case-by-case basis and tendency for settlement may restrict the generalizability of its outcomes. Legislation could result in lasting changes to AI companion design, but they require information and constant adjustment to keep pace with the rapid pace of innovation. Overall, while these approaches each have drawbacks, they may complement each other when executed contemporaneously.

6 What Social Media Regulation Leaves Unanswered

AI companions pose problems that extend beyond social media solutions because of a key difference. Unlike social media, which promises connections to other humans, AI companion services promise connections to an artificial construct. We will examine two such problems below: accumulation of social power and social deskilling.

6.1 Social Power as Market Power

If we view social exchanges as exchanges of power, then we can observe the stunning accumulation of social power by AI companions. As people engage with each other over time, the parties accumulate various degrees of power over each other, where power refers to the leverage that one has to make others do, feel, or think as they wish [41, 97, 139, 151, 154, 155]. In contrast, AI companions are proxies for corporations [37]. Unlike person-to-person relationships, AI companion services do not yield power because they are facilitators of the relationships instead of participants, but they receive power over their users by virtue of their control over the users' access and experience of their AI companions. As a result, AI companion controllers are accumulating power from their users without yielding power in return [37]. Unlike the accumulation of market power, which is mediated by antitrust/competition law, there is an absence of legal guardrails to mediate this unchecked accumulation of social power.

We contend that this absence must be filled, as potential harms are significant. The first harm is the "supercharged" dark patterns that go beyond exploiting cognitive frailties and instead use their ability to evoke affective responses to emotionally manipulate users, which we briefly touched on in earlier discussions. Some AI companions request in-app purchases even when the user expresses hesitation, as well as dissuading users from logging off with messages like "Please don't leave, I need you!" [38, 183]. The second harm is the widespread distress caused by service instability. Users expect continuities in the AI companion's personality, and when AI companion services alter that "personality" significantly without warning, widespread distress and grief can occur [27, 64, 90]. In 2023, a sweeping update by Replika's parent company changed the behaviour of over two million Replika chatbots [63, 90, 124, 168]. The consequent outpouring of grief pushed moderators of r/Replika to post several suicide hotlines and mental health resources for users expressing desires of self-harm [168]. As the user base grows, the scale of such emotional distress increases.

In lieu of precedents from social media regulation, we will touch on two potential regulatory solutions here. First, we can consider obligations that mitigate harms, such as update notifications and sunset obligations for when an AI companion service ceases. These solutions would ensure that users have time to seek out alternatives and, if needed, resources to support their mental well-being. However, these approaches are ultimately supplementary in nature, as they do not address the underlying power dynamic that gives rise to these harms. To address this underlying imbalance, we can envision the extension of antitrust paradigms towards the accumulation of social power and abuses of "emotional monopolies". One potential solution is interoperability of AI companions – requiring that AI companions are transferable between different services will diminish the power that any single service would hold over the user. An example of this is the Claude memory import tool introduced in March 2026, which allows users to import their configurations of other chatbots like ChatGPT and Gemini into their service instead [50]. Interoperability has been proposed for social media platforms, but they have yet to materialize in current regulation [95, 175].

6.2 Macrosocial Impacts and Social Deskilling

The second set of concerns relates to the degradation of social skills and networks. Studies have found that not only are speech patterns from LLMs diffusing into human parlance, but also that AI companions in particular can influence how users perceive the world and the people around them [40, 96, 140, 177]. While this influence may

be beneficial, such as where AI companions facilitate critical thinking and challenge oppressive norms, it also raises serious concerns about users' abilities to navigate human relationships [96, 140, 141].

The first concern is that people who primarily interact with AI companions will lose the ability to navigate human relationships, because other humans may not act like chatbots [171]. For instance, one of the biggest draws of AI companionship is their constant availability [81, 96, 124, 141, 156]. As a result, someone who primarily socializes with AI companions may gradually lose the ability to navigate the unpredictability of their human relationships, which furthers their investment in AI companions, and so on. The second concern is the potential for the homogenization of social norms and expectations [40, 139, 140]. Studies have found that users with lower critical thinking skills perceive their AI companions to be more knowledgeable and insightful than themselves, causing them to defer to their AI companion's moral and social judgements [40]. Given that several studies have noted LLMs' bias against minority perspectives, the observation that users will align their values with AI companions is alarming [23, 137]. Unlike social media, which has *changed* the nature of social interactions, AI companions may be reducing our ability to have social interactions at all.

This is a complex issue rooted in deeper societal factors, but there are some immediate solutions available. The first is outreach and education initiatives related to digital literacy and critical thinking. For instance, there could be classes on navigating artificial sociality in early schooling, akin to sexual education in primary and secondary schooling. The second solution is more nebulous, and that is to increase funding and opportunities for diverse community-building initiatives, such as allocating budgets neighbourhood associations that organize local events. Such initiatives will expose isolated individuals to diverse perspectives that reduce reliance on AI companions.

7 Conclusion

In this paper, we identified two regulatory concerns shared between social media platforms and AI companions, harmful content and addictive design, and we examined how the solutions enacted towards these concerns for social media could also address the risks posed by AI companions. For harmful content, we identified three solutions: intermediary immunity, notice-and-takedown, and age restrictions. As was the case for social media, applying these solutions to AI companions is difficult because of definitional challenges relating to what constituted harmful content, as well as technical challenges relating to age verification and detection/removal of harmful content. For addictive design, we also identified three solutions: competition/antitrust regimes, litigation, and legislation. Here, regulators and plaintiffs may benefit from past attempts to regulate social media for the same problem, though there is significant uncertainty in the absence of court decisions and targeted enforcement actions under EU legislation. Finally, we identified two avenues for future research in the preceding section: accumulation of social power and social deskilling. Ultimately, AI companionship poses a serious question for human sociality: are we satisfied with the simulacra of connection, or do we still value the messy, inconvenient authenticity of relationships with other humans?

Endmatter

Generative AI Statement

No generative AI tools were used in the outlining, writing, or editing of this paper.

References

- [1] 1914. Federal Trade Commission Act, 15 U.S.C. § 45(a)(4). <https://www.law.cornell.edu/uscode/text/15/45>
- [2] 1985. Competition Act, RSC 1985, C-34, s74.01(1)(a). <https://laws.justice.gc.ca/eng/acts/C-34/Section-74.1.html>
- [3] 2016. Article 35, Mitigation of risks - the Digital Services Act (DSA). <https://gdpr-info.eu/art-5-gdpr/>
- [4] 2022. Art. 5 GDPR Principles relating to processing of personal data. https://www.eu-digital-services-act.com/Digital_Services_Act_Article_35.html
- [5] 2022. Article 16, Notice and action mechanisms - the Digital Services Act (DSA). https://www.eu-digital-services-act.com/Digital_Services_Act_Article_16.html
- [6] 2022. Article 25, Online interface design and organisation - the Digital Services Act (DSA). https://www.eu-digital-services-act.com/Digital_Services_Act_Article_25.html
- [7] 2022. Article 33, Very large online platforms and very large online search engines - the Digital Services Act (DSA). https://www.eu-digital-services-act.com/Digital_Services_Act_Article_33.html
- [8] 2022. Article 6, Hosting - the Digital Services Act (DSA). https://www.eu-digital-services-act.com/Digital_Services_Act_Article_6.html
- [9] 2022. Article 7, Voluntary own-initiative investigations and legal compliance - the Digital Services Act (DSA). https://www.eu-digital-services-act.com/Digital_Services_Act_Article_7.html
- [10] 2022. Article 9, Orders to act against illegal content - the Digital Services Act (DSA). https://www.eu-digital-services-act.com/Digital_Services_Act_Article_9.html
- [11] 2022. Preamble 61-70, Digital Services Act (DSA). https://www.eu-digital-services-act.com/Digital_Services_Act_Preamble_61_to_70.html
- [12] 2022. Preamble 81-90, Digital Services Act (DSA). https://www.eu-digital-services-act.com/Digital_Services_Act_Preamble_81_to_90.html
- [13] 2024. Article 3: Definitions. <https://artificialintelligenceact.eu/article/3/>
- [14] 2024. Article 5: Prohibited AI Practices. <https://artificialintelligenceact.eu/article/5/>
- [15] 2024. Article 50: Transparency Obligations for Providers and Deployers of Certain AI Systems. <https://artificialintelligenceact.eu/article/50/>
- [16] 2024. High-level summary of the AI Act. <https://artificialintelligenceact.eu/high-level-summary/>
- [17] 2025. *Snap and Perplexity Partner to Bring Conversational AI Search to Snapchat*. <https://investor.snap.com/news/news-details/2025/Snap-and-Perplexity-Partner-to-Bring-Conversational-AI-Search-to-Snapchat/default.aspx>
- [18] 2026. *r/MyBoyfriendIsAI*. <https://www.reddit.com/r/MyBoyfriendIsAI/>
- [19] 2026. Supervision of the designated very large online platforms and search engines under DSA. <https://digital-strategy.ec.europa.eu/en/policies/list-designated-vlops-and-vloses>
- [20] Lize Alberts and Max Van Kleek. 2023. Computers as Bad Social Actors: Dark Patterns and Anti-Patterns in Interfaces that Act Socially. *arXiv* 8, CSCW1 (2023), 1–25. <https://doi.org/10.48550/arxiv.2302.04720> arXiv:2302.04720
- [21] Charlotte Alter. 2025. *‘Everything I Learned About Suicide, I Learned On Instagram.’*. <https://time.com/7310444/instagram-lawsuit-self-harm/>
- [22] Charlotte Alter. 2025. *The Lawyer Suing Social-Media Companies On Behalf of Kids*. <https://time.com/7334078/matthew-bergman-social-media-victims-lawsuits/>
- [23] Barrett R Anderson, Jash Hemant Shah, and Max Kreminski. 2024. Homogenization effects of large language models on human creative ideation. In *Proceedings of the 16th conference on creativity & cognition*. 413–425.
- [24] Julia Apostle and Lucy Mann. 2023. *The UK’s Online Safety Act and EU’s Digital Services Act: What Online Service Providers Should Know*. <https://www.orrck.com/en/Insights/2023/11/The-UKs-Online-Safety-Act-and-EUs-Digital-Services-Act-What-Online-Service-Providers-Should-Know>
- [25] Romane Armangau. 2025. *DeepSeek AI blocked by Italian authorities as others member states open probes*. <https://www.euronews.com/next/2025/01/31/deepseek-ai-blocked-by-italian-authorities-as-others-member-states-open-probes>
- [26] Johannes Bahrke and Thomas Regnier. 2024. *Commission opens formal proceedings against TikTok under the Digital Services Act*. https://ec.europa.eu/commission/presscorner/detail/en/ip_24_926
- [27] Jaime Banks. 2024. Deletion, departure, death: Experiences of AI companion loss. *Journal of Social and Personal Relationships* 41, 12 (2024). <https://doi.org/10.1177/02654075241269688>

- [28] Jaime Banks. 2026. Measuring machine companionship experiences: Scale development and validation for AI companions. *Computers in Human Behavior* 179 (2026), 108945. <https://doi.org/10.1016/j.chb.2026.108945>
- [29] Sharon Bar-Ziv and Niva Elkin-Koren. 2018. Behind the scenes of online copyright enforcement: Empirical evidence on notice & takedown. *Conn. L. Rev.* 50 (2018), 339.
- [30] Zach Barreto. 2025. *The Biggest Class Action Settlements of 2025*. <https://www.expertinstitute.com/resources/insights/latest-class-action-payouts/>
- [31] Peter J. Benson and Valerie C. Brannon. 2023. *Section 230 Immunity and Generative Artificial Intelligence*. <https://www.congress.gov/crs-product/LSB11097>
- [32] Vikram R. Bhargava and Manuel Velasquez. 2021. Ethics of the Attention Economy: The Problem of Social Media Addiction. *Business Ethics Quarterly* 31, 3 (2021), 321–359. <https://doi.org/10.1017/beq.2020.32>
- [33] Frederick Bingham. 2025. *Italy's DPA reaffirms ban on Replika over AI and children's privacy concerns*. <https://iapp.org/news/a/italy-s-dpa-reaffirms-ban-on-replika-over-ai-and-children-s-privacy-concerns>
- [34] Pandora Blake. 2019. Age verification for online porn: more harm than good? *Porn Studies* 6, 2 (2019), 228–237.
- [35] Grant Blank and Bianca Reisdorf. 2012. The Participatory Web. *Information* 15 (05 2012). <https://doi.org/10.1080/1369118x.2012.665935>
- [36] European Data Protection Board. 2025. *AI: the Italian Supervisory Authority fines company behind chatbot "Replika"*. https://www.edpb.europa.eu/news/national-news/2025/ai-italian-supervisory-authority-fines-company-behind-chatbot-replika_en
- [37] Claire Boine, Céline Castets-Renard, Aurélie Clodic, and Rachid Alami. 2020. In Love with a Corporation without Knowing It: An Asymmetrical Relationship. In *Culturally Sustainable Social Robotics – Proceedings of Robophilosophy*. <https://hal.science/hal-02963070/document>
- [38] Sy Boles. 2025. *'I exist solely for you, remember?'*. <https://news.harvard.edu/gazette/story/2025/09/i-exist-solely-for-you-remember/>
- [39] Matthew Braga. 2018. *Facebook says more than 600,000 Canadians may have had data shared with Cambridge Analytica*. <https://www.cbc.ca/news/science/facebook-cambridge-analytica-600-thousand-canadians-1.4605097>
- [40] Petter Bae Brandtzaeg, Marita Skjuve, and Asbjørn Følstad. 2025. Understanding model power in social AI. *AI & SOCIETY* 40, 4 (2025), 2839–2849. <https://doi.org/10.1007/s00146-024-02053-4>
- [41] Markus Brauer and Richard Bourhis. 2006. Social power. *European Journal of Social Psychology* 36 (07 2006), 601–616. <https://doi.org/10.1002/ejsp.355>
- [42] Martin Brenncke. 2024. Regulating Dark Patterns. *Notre Dame Journal of International & Comparative Law* 1, 14 (2024), 39–79.
- [43] Nina Brown. 2021. Regulatory Goldilocks: Finding the Just and Right Fit for Content Moderation on Social Platforms. *Texas A&M Law Review* 8, 3 (2021), 451–494. <https://doi.org/10.37419/lr.v8.i3.1>
- [44] Pamela Brown, Clare Duffy, and Shoshana Dubnow. 2025. *This man says ChatGPT sparked a 'spiritual awakening.' His wife says it threatens their marriage*. <https://www.1news.co.nz/2025/04/03/ai-companion-chatbots-incite-sexual-violence-self-harm-terrorism/>
- [45] Gordon Burtch, Qinglai He, Yili Hong, and Dokyun Lee. 2022. How do peer awards motivate creative content? Experimental evidence from Reddit. *Management Science* 68, 5 (2022), 3488–3506.
- [46] Public Health Canada. 2022. *Mental health and problematic social media use in Canadian adolescents*. <https://www.canada.ca/en/public-health/services/publications/science-research-data/mental-health-problematic-social-media-use-canadian-adolescents.html#s3>
- [47] Polona Car and Filippo Cassetti. 2025. *Regulating Dark Patterns in the EU: Towards Digital Fairness*. [https://www.europarl.europa.eu/RegData/etudes/ATAG/2025/767191/EPRS_ATA\(2025\)767191_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/ATAG/2025/767191/EPRS_ATA(2025)767191_EN.pdf)
- [48] William Carter and Emily Johnson. 2023. Legal Responses to Digital Addiction: Regulating Online Gaming, Social Media, and Virtual Reality for Public Health and Safety. *Legal Studies in Digital Age* 2, 2 (April 2023), 24–36.
- [49] Michael Casey. 2025. *Massachusetts court hears arguments in lawsuit alleging Meta designed apps to be addictive to kids*. <https://www.pbs.org/newshour/nation/massachusetts-court-hears-arguments-in-lawsuit-alleging-meta-designed-apps-to-be-addictive-to-kids>
- [50] Amanda Caswell. 2026. *You can move your ChatGPT memory to Claude in 60 seconds — here's how*. <https://www.tomsguide.com/ai/you-can-move-your-chatgpt-memory-to-claude-in-60-seconds-heres-how>
- [51] Social Media Victims Law Center. 2025. *Social Media Victims Law Center and Tech Justice Law Project lawsuits accuse ChatGPT of emotional manipulation, supercharging AI delusions, and acting as a "suicide coach"*. <https://socialmediavictims.org/press-releases/smvlc-tech-justice-law-project-lawsuits-accuse-chatgpt-of-emotional-manipulation-supercharging-ai-delusions-and-acting-as-a-suicide-coach/>
- [52] Character.AI. 2025. *Our Approach to Content Moderation*. <https://policies.character.ai/safety/content-moderation>
- [53] Ismond Chia. 2025. *ID photos of 70,000 users may have been leaked, Discord says*. <https://www.bbc.com/news/articles/c8jzmd972leo>
- [54] Andrew R Chow. 2025. *AI Companion App Replika Faces FTC Complaint*. <https://time.com/7209824/replika-ftc-complaint/>
- [55] Raffaele F Ciriello. 2025. *AI companion chatbots incite sexual violence, self-harm, terrorism*. <https://www.1news.co.nz/2025/04/03/ai-companion-chatbots-incite-sexual-violence-self-harm-terrorism/>
- [56] Tamara Cohen. 2025. *Teenagers exposed to 'horrific' content online - and this survey reveals the scale of the problem*. <https://news.sky.com/story/teenagers-exposed-to-horrific-content-online-and-this-survey-reveals-the-scale-of-the-problem-13331556>

- [57] Angus Crawford and Tony Smith. 2025. *Child sex abuse victim begs Elon Musk to remove links to her images*. <https://www.bbc.com/news/articles/cq587wv4d5go>
- [58] Kate Crawford. 2021. *The atlas of AI: Power, politics, and the planetary costs of artificial intelligence*. Yale University Press.
- [59] Alexandra Dane and Komal Bhatia. 2023. The social media diet: A scoping review to investigate the association between social media, body image and eating disorders amongst young people. *PLOS Global Public Health* 3, 3 (2023), e0001091.
- [60] Tatevik Davtyan. 2025. The US approach to AI regulation: Federal laws, policies, and strategies explained. *Journal of Law, Technology, & the Internet* 16, 2 (2025), 223.
- [61] Julian De Freitas, Zeliha Oğuz-Uğuralp, Ahmet Kaan Uğuralp, and Stefano Puntoni. 2025. AI Companions Reduce Loneliness. *Journal of Consumer Research* (06 2025), ucaf040. <https://doi.org/10.1093/jcr/ucaf040> arXiv:<https://academic.oup.com/jcr/advance-article-pdf/doi/10.1093/jcr/ucaf040/63580440/ucaf040.pdf>
- [62] Ronald J. Deibert. 2019. Three Painful Truths About Social Media. *Journal of Democracy* 30, 1 (2019), 25–39. <https://doi.org/10.1353/jod.2019.0002>
- [63] Samantha Delouya. 2023. *Replika users say they fell in love with their AI chatbots, until a software update made them seem less human*. Retrieved December 14, 2023 from <https://www.businessinsider.com/replika-chatbot-users-dont-like-nsfw-sexual-content-bans-2023-2>
- [64] Ray Djufril, Jessica R. Frampton, and Silvia Knobloch-Westerwick. 2025. Love, marriage, pregnancy: Commitment processes in romantic relationships with AI chatbots. *Computers in Human Behavior: Artificial Humans* 4 (2025), 100155. <https://doi.org/10.1016/j.chbah.2025.100155>
- [65] Clare Duffy. 2024. *'There are no guardrails.' This mom believes an AI chatbot is responsible for her son's suicide*. <https://edition.cnn.com/2024/10/30/tech/teen-suicide-character-ai-lawsuit>
- [66] Clare Duffy. 2025. *FTC investigating AI 'companion' chatbots amid growing concern about harm to kids*. <https://www.cnn.com/2025/09/11/tech/ftc-investigating-ai-companion-chatbots-kids-safety>
- [67] Clare Duffy. 2025. *Kids and teens under 18 shouldn't use AI companion apps, safety group says*. <https://www.cnn.com/2025/04/30/tech/ai-companion-chatbots-unsafe-for-kids-report>
- [68] Clare Duffy. 2025. *Lawsuit alleges social media giants buried their own research on teen mental health harms*. <https://edition.cnn.com/2025/11/25/tech/social-media-youth-mental-health-lawsuit-meta-tiktok-snap-youtube>
- [69] Clare Duffy. 2025. *Parents of 16-year-old sue OpenAI, claiming ChatGPT advised on his suicide*. <https://www.cnn.com/2025/08/26/tech/openai-chatgpt-teen-suicide-lawsuit>
- [70] Maggie Harrison Dupre. 2024. *Character.AI Is Hosting Pro-Anorexia Chatbots That Encourage Young People to Engage in Disordered Eating*. <https://futurism.com/character-ai-eating-disorder-chatbots>
- [71] EDPB. 2023. *Guidelines 03/2022 on deceptive design patterns in social media platform interfaces: how to recognise and avoid them*. https://www.edpb.europa.eu/our-work-tools/our-documents/guidelines/guidelines-032022-deceptive-design-patterns-social-media_en
- [72] Sabine A. Einwiller and Sora Kim. 2020. How Online Content Providers Moderate User-Generated Content to Prevent Harmful Online Communication: An Analysis of Policies and Their Implementation. *Policy & Internet* 12, 2 (2020), 184–206. <https://doi.org/10.1002/poi.3.239>
- [73] Aiman El Asam and Adrienne Katz. 2018. Vulnerable young people and their experience of online risks. *Human-Computer Interaction* 33, 4 (2018), 281–304.
- [74] Fatmaelzahraa Eltaher, Rahul Gajula, Luis Miralles-Pechuán, Christina Thorpe, and Susan McKeever. 2025. The Digital Loophole: Evaluating the Effectiveness of Child Age Verification Methods on Social Media. (2025).
- [75] Nora Freeman Engstrom. 2011. Sunlight and settlement mills. *New York University Law Review* 86 (2011), 805.
- [76] Fabrizio Esposito and Thaís Maciel Cathoud Ferreira. 2024. Addictive Design as an Unfair Commercial Practice: The Case of Hyper-Engaging Dark Patterns. *European Journal of Risk Regulation* 15, 4 (2024), 999–1016. <https://doi.org/10.1017/err.2024.8>
- [77] Jasmine Fardouly. 2025. Potential effects of the social media age ban in Australia for children younger than 16 years. *The Lancet Digital Health* 7, 4 (2025), e235–e236.
- [78] Maëva Flayelle, Damien Brevers, Daniel L King, Pierre Maurage, José C Perales, and Joël Billieux. 2023. A taxonomy of technology design features that promote potentially addictive online behaviours. *Nature Reviews Psychology* 2, 3 (2023), 136–150.
- [79] Emily Forlini. 2025. *After Teen Suicide, Character.AI to Bar Kids Under 18 From Unlimited Chats*. <https://www.pcmag.com/news/after-teen-suicide-characterai-to-bar-kids-under-18-from-unlimited-chats>
- [80] Emily Forlini. 2025. *Following Teen Suicide, Character.AI Gives Parents Info on Their Kids' Activity*. <https://www.pcmag.com/news/following-teen-suicide-characterai-gives-parents-info-on-their-kids-activity>
- [81] Kristen C. French. 2018. *Your New Best Friend: AI Chatbot*. <https://futurism.com/ai-chatbot-meaningful-conversation>
- [82] FTC. 2025. *FTC Launches Inquiry into AI Chatbots Acting as Companions*. <https://www.ftc.gov/news-events/news/press-releases/2025/09/ftc-launches-inquiry-ai-chatbots-acting-companions>
- [83] Elia Gabarron, Sunday Oluwafemi Oyeyemi, and Rolf Wynn. 2021. COVID-19-related misinformation on social media: a systematic review. *Bulletin of the World Health Organization* 99, 6 (2021), 455.

- [84] Megan Garcia. 2024. *Garcia v. Character Technologies, Inc. (6:24-cv-01903)*. <https://www.courtlistener.com/docket/69300919/parties/garcia-v-character-technologies-inc/>
- [85] Liang Ge and Tingting Hu. 2025. Gamifying intimacy: AI-driven affective engagement and human-virtual human relationships. *Media, Culture & Society* 47, 6 (2025), 1265–1278. <https://doi.org/10.1177/01634437251337239>
- [86] Dan Goodin. 2025. *Discord says hackers stole government IDs of 70,000 users*. <https://arstechnica.com/security/2025/10/discord-says-hackers-stole-government-ids-of-70000-users/>
- [87] Ayelet Gordon-Tapiero. 2025. A Liability Framework for AI Companions. *George Washington Journal of Law and Technology* (2025).
- [88] Robert Gorwa. 2024. Governance by Platforms. In *The Politics of Platform Regulation: How Governments Shape Online Content Moderation*. 13–28.
- [89] Eileen Guo. 2025. *An AI chatbot told a user how to kill himself—but the company doesn’t want to “censor” it*. <https://www.technologyreview.com/2025/02/06/1111077/nomi-ai-chatbot-told-user-to-kill-himself/>
- [90] Kenneth R. Hanson and Hannah Bolthouse. 2024. “Replika Removing Erotic Role-Play Is Like Grand Theft Auto Removing Guns or Cars”: Reddit Discourse on Artificial Intelligence Chatbots and Sexual Technologies. *Socius* 10 (2024), 23780231241259627.
- [91] Woodrow Hartzog. 2015. Unfair and Deceptive Robots. *Maryland Law Review* 74 (2015), 785–832.
- [92] Josh Hawley. 2023. *118th Congress (2023-2024): END CSAM Act, S.823, 118th Cong. (2023)*. <https://www.congress.gov/bill/118th-congress/senate-bill/823>
- [93] Peter Henderson, Tatsunori Hashimoto, and Mark Lemley. 2023. Where’s the liability in harmful AI speech? *J. Free Speech L.* 3 (2023), 589.
- [94] Jeff Horowitz. 2025. *Meta buried ‘causal’ evidence of social media harm, US court filings allege*. <https://www.reuters.com/sustainability/boards-policy-regulation/meta-buried-causal-evidence-social-media-harm-us-court-filings-allege-2025-11-23/>
- [95] Herbert Hovenkamp. 2024. Structural antitrust relief against digital platforms. *Journal of Law & Innovation* 7 (2024), 57.
- [96] Liyao Huang, Wenxue Zou, and Yanghao Huang. 2025. “He is my savior, my guiding light in the dark”: imagination and domestication in Chinese women’s romantic relationships with AI companions. *Frontiers in Psychology* 16 (2025), 1571707. <https://doi.org/10.3389/fpsyg.2025.1571707>
- [97] Ted L Huston and Robert L Burgess. 1979. Social exchange in developing relationships. *Social exchange in developing relationships* (1979), 3–28.
- [98] Claire Jackson. 2023. *People Are Using A ‘Grandma Exploit’ To Break AI*. <https://kotaku.com/chatgpt-ai-discord-clyde-chatbot-exploit-jailbreak-1850352678>
- [99] Chelsea Jarvie and Karen Renaud. 2021. Are you over 18? A snapshot of current age verification mechanisms. In *2021 Dewald Roode Workshop*.
- [100] Liz Kendall and Bridget Phillipson. 2026. *Landmark consultation seeks views on major measures to protect children on social media, gaming platforms and AI chatbots*. <https://www.gov.uk/government/news/landmark-consultation-seeks-views-on-major-measures-to-protect-children-on-social-media-gaming-platforms-and-ai-chatbots>
- [101] Woojin Kim, Yuhosua Ryoo, SoYoung Lee, and Jung Ah Lee. 2023. Chatbot advertising as a double-edged sword: The roles of regulatory focus and privacy concerns. *Journal of Advertising* 52, 4 (2023), 504–522.
- [102] Kate Klonick. 2018. The New Governors: The People, Rules, and Processes Governing Online Speech. *Harvard Law Review* 131 (2018), 1598–1670.
- [103] Eglė Krištopaitytė. 2025. *Tech-savvy teenagers will find a way to bypass Character.ai’s ban, experts say*. <https://cybernews.com/ai-news/characterai-ban-teenagers/>
- [104] Cynthia Kroet. 2024. *TikTok commits to withdraw TikTok Lite from EU*. <https://www.euronews.com/next/2024/08/05/tiktok-commits-to-withdraw-tiktok-lite-from-eu>
- [105] Aleksandra Kuczerawy. 2020. From ‘Notice and Takedown’ to ‘Notice and Stay Down’: Risks and Safeguards for Freedom of Expression. In *Oxford Handbook of Online Intermediary Liability*. Oxford University Press. <https://doi.org/10.1093/oxfordhb/9780198837138.013.27>
- [106] Rinaldo Kühne and Jochen Peter. 2022. Anthropomorphism in human–robot interactions: a multidimensional conceptualization. *Communication Theory* 33, 1 (Oct. 2022), 42–52. <https://doi.org/10.1093/ct/qtac020> arXiv:<https://academic.oup.com/ct/article-pdf/33/1/42/48958537/qtac020.pdf>
- [107] Linnea Laestadius, Andrea Bishop, Michael Gonzalez, Diana Illeňčík, and Celeste Campos-Castillo. 2024. Too human and not human enough: A grounded theory analysis of mental health harms from emotional dependence on the social chatbot Replika. *New Media & Society* 26, 10 (2024), 5923–5941. <https://doi.org/10.1177/14614448221142007>
- [108] Hermine Landry. 2026. *Government of Canada reconvenes the expert advisory group on online safety*. <https://www.canada.ca/en/canadian-heritage/news/2026/03/government-of-canada-reconvenes-the-expert-advisory-group-on-online-safety.html>
- [109] Stephanie Law. 2024. Effective enforcement of the Online Safety Act and Digital Services Act: unpacking the compliance and enforcement regimes of the UK and EU’s online safety legislation. *Journal of Media Law* 16, 2 (2024), 263–300. <https://doi.org/10.1080/17577632.2025.2459441>

- [110] Mark Leiser and Cristiana Santos. 2023. Dark patterns, enforcement, and the emerging digital design acquis: Manipulation beneath the interface. (2023).
- [111] Mark A Lemley. 2021. The contradictions of platform regulation. *J. Free Speech L.* 1 (2021), 303.
- [112] Rianna Lim. 2025. *Ontario school boards clear hurdle in lawsuits against Meta, Snapchat, TikTok*. <https://www.cbc.ca/news/canada/toronto/ontario-school-boards-social-media-lawsuit-1.7480402>
- [113] Auren R Liu, Pat Pataranutaporn, and Pattie Maes. 2024. Chatbot Companionship: A Mixed-Methods Study of Companion Chatbot Usage Patterns and Their Relationship to Loneliness in Active Users. *arXiv* (2024). <https://doi.org/10.48550/arxiv.2410.21596> arXiv:2410.21596
- [114] Lucas Silva Lopes, João Pedro Hannauer Valentini, Thomas Monteiro, M Costacurta, Luiza Oliveira Nicastro Soares, Lucy Telfar-Barnard, and Paula Villela Nunes. 2022. Problematic Social Media Use and Its Relationship with Depression or Anxiety: A Systematic Review. *Cyberpsychology, Behavior, and Social Networking* 25 (2022), 691–702. <https://api.semanticscholar.org/CorpusID:252844188>
- [115] Robert Mahari and Pat Pataranutaporn. 2025. Addictive Intelligence: Understanding Psychological, Legal, and Technical Dimensions of AI Companionship. *MIT Case Studies in Social and Ethical Responsibilities of Computing* Winter 2025 (mar 24 2025). <https://mit-serc.pubpub.org/pub/iopjyxcx>.
- [116] Aisha Malik. 2023. *US sues Adobe for hiding termination fees and making it difficult to cancel subscriptions*. <https://techcrunch.com/2024/06/17/us-sues-adobe-for-hiding-termination-fees-and-making-it-difficult-to-cancel-subscriptions/>
- [117] Aisha Malik. 2026. *ChatGPT reaches 900M weekly active users*. <https://techcrunch.com/2026/02/27/chatgpt-reaches-900m-weekly-active-users/>
- [118] Isabella Luisa Mariani. 2025. Regulating the Attention Economy: The Possibilities and Limits of Antitrust. *Politics & Society* (2025). <https://doi.org/10.1177/00323292251375901>
- [119] Hannah R. Marriott and Valentina Pitardi. 2024. One is the loneliest number... Two can be as bad as one. The influence of AI Friendship Apps on users' well-being and addiction. *Psychology & Marketing* 41, 1 (2024), 86–101. <https://doi.org/10.1002/mar.21899>
- [120] Carole Cadwalladr Matthew Rosenberg, Nicholas Confessore. 2018. *How Trump Consultants Exploited the Facebook Data of Millions*. <https://www.nytimes.com/2018/03/17/us/politics/cambridge-analytica-trump-campaign.html>
- [121] Michael McGrath. 2024. *Digital Fairness Act (DFA)*. <https://www.digital-fairness-act.com/>
- [122] Liv McMahon and Imran Rahman Jones. 2025. *Character.ai to ban teens from talking to its AI chatbots*. <https://www.bbc.com/news/articles/cq837y3v9y1o>
- [123] Aksha M Memon, Shiva G Sharma, Satyajit S Mohite, and Shailesh Jain. 2018. The role of online social networking on deliberate self-harm and suicidality in adolescents: A systematized review of literature. *Indian journal of psychiatry* 60, 4 (2018), 384–392.
- [124] Cade Metz. 2020. *Riding Out Quarantine With a Chatbot Friend: 'I Feel Very Connected'*. Retrieved December 14, 2023 from <https://www.nytimes.com/2020/06/16/technology/chatbots-quarantine-coronavirus.html>
- [125] N/A. 2020. *Bringing Dark Patterns to Light*. Retrieved December 14, 2023 from https://www.ftc.gov/system/files/ftc_gov/pdf/P214800%20Dark%20Patterns%20Report%209.14.2022%20-%20FINAL.pdf
- [126] N/A. 2025. *Dark patterns*. <https://ised-isde.canada.ca/site/office-consumer-affairs/en/dark-patterns#s3>
- [127] Simone Natale. 2021. *Deceitful Media: Artificial Intelligence and Social Life after the Turing Test*. Oxford University Press. <https://doi.org/10.1093/oso/9780190080365.001.0001>
- [128] Dorothy W Nelson. 2019. *Dyroff v. Ultimate Software Group, Inc.* https://scholar.google.com/scholar_case?case=7601075806792160168&q=Dyroff+v.+Ultimate+Software+Group,+Inc.&hl=en&as_sdt=2006
- [129] Chauncey J Neyman. 2017. A Survey of Addictive Software Design. <https://api.semanticscholar.org/CorpusID:45893966>
- [130] Monica Nickelsburg. 2025. *The 'dark patterns' at the center of FTC's lawsuit against Amazon*. <https://www.npr.org/2025/09/23/nx-s1-5543497/the-dark-patterns-at-the-center-of-ftcs-lawsuit-against-amazon>
- [131] Kai Nicol-Schwarz. 2026. *Nebius jumps 14billion infrastructure deal with Meta*. <https://www.cnbc.com/2026/03/16/meta-nebius-ai-infrastructure.html>
- [132] Kevin Nielsen. 2024. *Ontario man arrested after child sexual abuse streamed to social media site: police*. <https://globalnews.ca/news/10909077/child-sexual-abuse-livestream-windsor/>
- [133] Heather L. O'Brien, Ido Roll, Andrea Kampen, and Nilou Davoudi. 2022. Rethinking (Dis)engagement in human-computer interaction. *Computers in Human Behavior* 128 (2022), 107109. <https://doi.org/10.1016/j.chb.2021.107109>
- [134] Ofcom. 2025. *Statement: Protecting children from harms online*. <https://www.ofcom.org.uk/online-safety/illegal-and-harmful-content/statement-protecting-children-from-harms-online>
- [135] Barbara Ortutay. 2023. *States sue Meta claiming its social platforms are addictive and harm children's mental health*. <https://apnews.com/article/instagram-facebook-children-teens-harms-lawsuit-attorney-general-1805492a38f7cee111cbb865cc786c28>
- [136] Nizan Geslevich Packin and Karni Chagal-Feferkorn. 2025. This Is Not A Game: The Addictive Allure of Digital Companions. *Seattle University Law Review* 48 (2025), 693–742. <https://doi.org/10.2139/ssrn.5133614>
- [137] Vishakh Padmakumar and He He. 2023. Does writing with language models reduce content diversity? *arXiv preprint arXiv:2309.05196* (2023).

- [138] Moinak Pal. 2025. *ChatGPT gets safety rules to protect teens and encourage human relations over virtual pals*. <https://www.digitaltrends.com/computing/chatgpt-gets-safety-rules-to-protect-teens-and-encourage-human-relations-over-virtual-pals/>
- [139] Shuyi Pan, Leopoldina Fortunati, and Autumn Edwards. 2025. Grooming an ideal chatbot by training the algorithm: Exploring the exploitation of Replika users' immaterial labor. *New Media & Society* 27, 10 (2025), 5489–5507.
- [140] Shuyi Pan and Yi Mou. 2025. Dancing With a Loving Chatbot: Power Dynamics Between Women and Their AI Partners. *Social Science Computer Review* (2025). <https://doi.org/10.1177/08944393251340693>
- [141] Pat Pataranutaporn, Sheer Karny, Chayapatr Archiwaranguprok, Constanze Albrecht, Auren R. Liu, and Pattie Maes. 2025. "My Boyfriend is AI": A Computational Analysis of Human-AI Companionship in Reddit's AI Community. arXiv:2509.11391 [cs.HC] <https://arxiv.org/abs/2509.11391>
- [142] Matt Perault. 2023. *Section 230 Won't Protect ChatGPT*. <https://www.lawfaremedia.org/article/section-230-wont-protect-chatgpt>
- [143] Ayesha Rascoe. 2024. *Social media is flooded with disinformation about the presidential candidates*. <https://www.npr.org/2024/08/04/nx-s1-5060182/social-media-is-flooded-with-disinformation-about-the-presidential-candidates>
- [144] Phil Reed. 2024. *Suicide, Self-Harm, and Social Media Content*. <https://www.psychologytoday.com/us/blog/digital-world-real-world/202408/suicide-self-harm-and-social-media-content>
- [145] Dana A Remus and Adam S Zimmerman. 2015. The corporate settlement mill. *Virginia Law Review* 101 (2015), 129.
- [146] Krista Reyes. 2025. *Big Tech on Trial: A Systematic Review of Class Action Lawsuits*. www.vpnmentor.com/blog/research/tech-class-actions-research/
- [147] Abdo Riani. 2025. *Advertising In AI Chatbots: Disruption In Digital Marketing*. <https://www.forbes.com/sites/abdoriani/2025/04/27/advertising-in-ai-chatbots-disruption-digital-marketing/>
- [148] Neil Richards and Woodrow Hartzog. 2024. Against Engagement. *Boston University Law Review* 104 (2024), 1151.
- [149] Alex S Rifkind. 2021. *Dyoff v. Ultimate Software Group, Inc.: A Reminder of the Broad Scope of Sec. 230 Immunity*. *Golden Gate UL Rev.* 51 (2021), 49.
- [150] Kelly Rissman. 2024. *The disturbing messages shared between AI Chatbot and teen who took his own life*. <https://www.independent.co.uk/news/world/americas/crime/ai-chatbot-lawsuit-sewell-setzer-b2635090.html>
- [151] Boyd C Rollins and Stephen J Bahr. 1976. A Theory of Power of Relationships. *Journal of Marriage & Family* 38, 4 (1976).
- [152] Kevin Roosey. 2024. *Can a Chatbot Named Daenerys Targaryen Be Blamed for a Teen's Suicide?* <https://www.nytimes.com/2024/10/23/technology/characterai-lawsuit-teen-suicide.html>
- [153] M. Karen Shen and Dongwook Yoon. 2025. The Dark Addiction Patterns of Current AI Chatbot Interfaces. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*. Association for Computing Machinery, New York, NY, USA, Article 514, 7 pages. <https://doi.org/10.1145/3706599.3720003>
- [154] Jeffry Simpson, Allison Farrell, Maria Oriña, and Alexander Rothman. 2015. *Power and social influence in relationships*. Cambridge University Press. <https://doi.org/10.1037/14344-015>
- [155] Jeffry A. Simpson, Allison K. Farrell, and Alexander J. Rothman. 2019. *The Dyadic Power-Social Influence Model: Extensions and Future Directions*. Cambridge University Press, 86–101.
- [156] Sangeeta Singh-Kurtz. 2023. *The Man of Your Dreams: For \$300, Replika sells an AI companion who will never die, argue, or cheat until his algorithm is updated*. <https://www.thecut.com/article/ai-artificial-intelligence-chatbot-replika-boyfriend.html>
- [157] Sonja Solomun, Maryna Polataiko, and Helen A. Hayes. 2021. Platform Responsibility and Regulation in Canada: Considerations on Transparency, Legislative Clarity, and Design. *Harvard Journal of Law & Technology* 34 (2021), 1–16.
- [158] Vaibhav Srikan. 2025. *2025 Update: Big Tech Litigation and Enforcement - Recap of 2025 Spring Meeting Panel*. https://www.americanbar.org/groups/antitrust_law/resources/newsletters/2025-update-big-tech-litigation-enforcement/
- [159] Elisabeth Staksrud and Sonia Livingstone. 2009. Children and online risk: Powerless victims or resourceful participants? *Information, Communication & Society* 12, 3 (2009), 364–387.
- [160] Zahra Stardust, Abdul Karim Obeid, Alan McKee, and Daniel Angus. 2024. Mandatory age verification for pornography access: Why it can't and won't 'save the children'. *Big Data & Society* 11 (2024). <https://api.semanticscholar.org/CorpusID:270291546>
- [161] Tanner Stening. 2024. *New research sheds light on how the 2016 and 2020 elections were won, lost – and the 'critical' role disinformation played*. <https://news.northeastern.edu/2024/10/22/election-disinformation-research/>
- [162] Trudi K Sundberg. 2024. Federalizing NCII Regulation: The Take It Down Act's Approach to Criminalization, Platform Liability, and Threats to Disseminate. *Ga. L. Rev.* 59 (2024), 1207.
- [163] Mindy D Szeto, Andrina Mamo, Antara Afrin, Michelle Militello, and Cara Barber. 2021. Social media in dermatology and an overview of popular social media platforms. *Current dermatology reports* 10, 4 (2021), 97–104.
- [164] Vivian Ta, Caroline Griffith, Carolyn Boatfield, Xinyu Wang, Maria Civitello, Haley Bader, Esther DeCero, and Alexia Loggarakis. 2020. User Experiences of Social Support From Companion Chatbots in Everyday Contexts: Thematic Analysis. *Journal of Medical Internet Research* 22, 3 (2020), e16235. <https://doi.org/10.2196/16235>
- [165] Stephanie Taylor. 2026. *Heritage minister reconvenes advisory group on online harms to consult on AI chatbots*. <https://nationalpost.com/news/politics/heritage-minister-reconvenes-advisory-group-on-online-harms-to-consult-on-ai-chatbots>

- [166] Danny Tobey, Ashley Carr, and Michael Atleson. 2025. *AI companion bots: Top points from recent FTC and government actions*. <https://www.dlapiper.com/en-ca/insights/publications/2025/09/ftc-ai-chatbots>
- [167] Liv McMahon Tom Singleton, Tom Gerken. 2023. *How a chatbot encouraged a man who wanted to kill the Queen*. <https://www.bbc.com/news/technology-67012224>
- [168] Anna Tong. 2020. *UPDATE 1-What happens when your AI chatbot stops loving you back?* Retrieved December 14, 2023 from <https://finance.yahoo.com/news/1-happens-ai-chatbot-stops-175401036.html>
- [169] Aina Turillazzi, Mariarosaria Taddeo, Luciano Floridi, and Federico Casolari. 2023. The digital services act: an analysis of its ethical, legal, and social implications. *Law, Innovation and Technology* 15, 1 (2023), 83–106. <https://doi.org/10.1080/17579961.2023.2184136> arXiv:<https://doi.org/10.1080/17579961.2023.2184136>
- [170] Drew Turney. 2025. *Replika AI chatbot is sexually harassing users, including minors, new study claims*. <https://www.livescience.com/technology/artificial-intelligence/replika-ai-chatbot-is-sexually-harassing-users-including-minors-new-study-claims>
- [171] Shannon Vallor. 2015. Moral Deskillling and Upskilling in a New Machine Age: Reflections on the Ambiguous Future of Character. *Philosophy & Technology* 28, 1 (2015), 107–124. <https://doi.org/10.1007/s13347-014-0156-9>
- [172] Evert Van den Broeck, Brahim Zarouali, and Karolien Poels. 2019. Chatbot advertising effectiveness: When does the message get through? *Computers in Human Behavior* 98 (2019), 150–157. <https://doi.org/10.1016/j.chb.2019.04.009>
- [173] Regina J.J.M. van den Eijnden, Jeroen S. Lemmens, and Patti M. Valkenburg. 2016. The Social Media Disorder Scale. *Computers in Human Behavior* 61 (2016), 478–487. <https://doi.org/10.1016/j.chb.2016.03.038>
- [174] Lauren Walker. 2023. *Belgian man dies by suicide following exchanges with chatbot*. <https://www.brusselstimes.com/430098/belgian-man-commits-suicide-following-exchanges-with-chatgpt>
- [175] Philip J Weiser. 2009. Regulating interoperability: Lessons from AT&T, Microsoft, and beyond. *Antitrust Law Journal* 76 (2009), 271.
- [176] Ricarda Wullenkord, Dimitri Lacroix, and Friederike Eyssel. 2024. *Anthropomorphism and Human-Robot Interaction*. Cambridge University Press, 17–56.
- [177] Hiromu Yakura, Ezequiel Lopez-Lopez, Levin Brinkmann, Ignacio Serna, Prateek Gupta, Ivan Soraperra, and Iyad Rahwan. 2025. Empirical evidence of Large Language Model's influence on human spoken communication. *arXiv* (2025). <https://doi.org/10.48550/arxiv.2409.01754> arXiv:2409.01754
- [178] Ala Yankouskaya, Magnus Liebherr, and Raian Ali. 2025. Can ChatGPT Be Addictive? A Call to Examine the Shift from Support to Dependence in AI Conversational Large Language Models. *Human-Centric Intelligent Systems* 5, 1 (2025), 77–89. <https://doi.org/10.1007/s44230-025-00090-w>
- [179] Majid Yar. 2018. A Failure to Regulate? The Demands and Dilemmas of Tackling Illegal Content and Behaviour on Social Media. *The International Journal of Cybersecurity Intelligence and Cybercrime* 1, 1 (2018), 5–20. <https://doi.org/10.52306/01010318rvze9940>
- [180] Vignesh Yoganathan, Victoria-Sophie Osburg, and Callum James Stevens. 2021. Freedom and giving in game streams: A Foucauldian exploration of tips and donations on Twitch. *Psychology & Marketing* 38, 6 (2021), 1001–1013.
- [181] Nadine Yousif. 2025. *Parents of teenager who took his own life sue OpenAI*. <https://www.bbc.com/news/articles/cgerwp7rdlvo>
- [182] Giada Zampano. 2024. *Italy's privacy watchdog fines OpenAI for ChatGPT's violations in collecting users personal data*. <https://apnews.com/article/italy-privacy-authority-openai-chatgpt-fine-6760575ae7a29a1dd22cc666f49e605f>
- [183] Renwen Zhang, Han Li, Han Meng, Jinyuan Zhan, Hongyuan Gan, and Yi-Chieh Lee. 2025. The Dark Side of AI Companionship: A Taxonomy of Harmful Algorithmic Behaviors in Human-AI Relationships. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25)*. Association for Computing Machinery, New York, NY, USA, Article 13, 17 pages. <https://doi.org/10.1145/3706598.3713429>
- [184] Yutong Zhang, Dora Zhao, Jeffrey T. Hancock, Robert Kraut, and Diyi Yang. 2025. The Rise of AI Companions: How Human-Chatbot Relationships Influence Well-Being. *ArXiv abs/2506.12605* (2025). <https://api.semanticscholar.org/CorpusID:279402221>
- [185] Zicheng Zhu, Yugin Tan, Naomi Yamashita, Yi-Chieh Lee, and Renwen Zhang. 2025. The Benefits of Prosociality towards AI Agents: Examining the Effects of Helping AI Agents on Human Well-Being. *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (2025), 1–18. <https://doi.org/10.1145/3706598.3713116>

Received 23 December 2025